MSE 26000: THERMODYNAMICS OF MATERIALS (CRN: 49580)
Spring Semester, 2026

Objective: To examine: the fundamental laws of thermodynamics and their applications to material systems, criteria for equilibrium, reaction and phase equilibria, properties of solutions, and thermodynamic origins of phase diagrams.

Prerequisites: MA 26100

Corequisites: MSE 23000, CHM 11600

Instructional Modality: Two face-to-face lectures, each of 75 minutes duration, per week (150 minutes of total lecture time per week).

Attendance: This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for University-sponsored activities and religious observations, the student should inform the instructor of the situation at least 2 weeks in advance. Procedures for other types of excused absences (e.g., for grief/bereavement, military service, jury duty, parenting leave, or certain medical care) can be found on the website for the Office of the Dean of Students (https://www.purdue.edu/advocacy/students/absence-policies.html). For unanticipated or emergency absences when advance notification is not possible, contact the instructor as soon as possible by Purdue e-mail or phone.

Office Hours: A single scheduled session of 1.5 hour duration, available to all of the students, will be held each week, provided that the instructor is notified of intended attendance by at least 1 student (with notification to the instructor in class or by e-mail) at least 24 h prior to the scheduled 1.5 hour session.

Main Textbook: David R. Gaskell, David E. Laughlin, *Introduction to the Thermodynamics of Materials*, 7th Edition, CRC Press, Taylor & Francis Group, 2024 (available at the Purdue University Bookstore, 360 W. State Street, W. Lafayette, IN)

Other References: (on reserve in WALC) *Chemical Thermodynamics of Materials*, C. H. P. Lupis, North Holland, 1983; *Thermodynamics, Kinetic Theory, and Statistical Thermodynamics*, Francis W. Sears, Gerhard L. Salinger, Addison-Wesley, 1984

Homework: Homework exercises (mostly from the main textbook), and associated solutions, will be given to augment the lectures and reading assignments. While the homework will not be graded, it is strongly recommended that students perform these homework exercises in an independent and timely manner to better understand the course material and to prepare for quizzes, the midterm exam, and the final exam.

Grading:

Quizzes:	30%
Midterm Exam:	30%
Final Exam:	40%
Total:	100%

Quizzes: Each quiz (closed) will be based on material covered in homework problems, in previous lectures, and/or in the assigned reading. Quiz dates will be announced ahead of time during a preceding class period, and each quiz will be held during the normal class period. No calculators, cell phones, laptops, smart watches, or any other electronic device (e.g., electronic storage or internet-accessible device) or associated applications (such as ChatGPT) are allowed to be used during a quiz. The use of notes and reference materials are also not allowed during a quiz. The lowest quiz grade for each student will be dropped, and the overall quiz grade for a given student will then be based on the average of other quiz scores for that student.

Midterm Exam: Unless otherwise indicated by the instructor, a closed midterm exam, of 75 minutes duration, will be held in the 8th week of the semester. No calculators, cell phones, laptops, smart watches, or any other electronic device (e.g., electronic storage or internet-accessible device) or associated applications (such as ChatGPT) are allowed to be used during this midterm exam. The use of notes and reference materials are also not allowed during this midterm exam.

Students will be given an opportunity for a "second attempt" of designated problems on this midterm exam at home (after the midterm exam is held). A second copy of the these designated problems will be provided to each student to take home, so that each student can individually solve such problems (with each student showing step-by-step solutions ***in their own handwriting/printing***, not typed). If an enhanced score is obtained on this second try effort (which must include step-by-step solutions by the student for the designated problems), then this enhanced score will be added to the midterm exam score for a possible increase of up to two partial letter grades (e.g., a B grade obtained on the midterm exam could be improved up to an A- at most; or a C+ grade on the midterm exam could be improved up to a B at most). Unless otherwise informed by the instructor, the deadline for the solutions to these second attempt problems will be 1 hour prior to the start of the second class following the midterm exam. No extensions will be allowed (i.e., solutions to the second attempt problems submitted after this deadline will not be graded). These second-attempt problem solutions may be submitted in person to the instructor in person at the instructor's office or via e-mail to the instructor as an attached single scanned word or pdf file (with the file name containing the name of the student).

Such second-attempt problem solving for the midterm exam is optional (note: because the letter grade scale will be set by midterm exam scores, any student who does not submit second-attempt solutions will not be penalized).

Final Exam: The (closed) final exam (2 hours) will be given in a to-be-designated room during the final exam week. No calculators, cell phones, laptops, smart watches, or any other electronic device (e.g., electronic storage or internet-accessible device) or associated applications (such as ChatGPT) are allowed to be used during the final exam. The use of notes and reference materials are also not allowed during the final exam.

Grading Scale: This class will be graded according to a curve based on comparative student performance; that is, final course grades will be distributed through a comparison of relative student performance on the quizzes, midterm exam, and final exam, according to the weighting of the scores for the quizzes, midterm exam, and final exam as outlined above.

Learning Outcomes: By the end of this course, you will be able to:

1. Apply the 1st, 2nd, and 3rd laws of thermodynamics for closed systems, along with Maxwell's equations and auxiliary functions, to understand equilibrium associated with single component systems (e.g., to evaluate changes in thermodynamic properties as a function of temperature, pressure, and volume; to correlate various thermodynamic properties to each other; to evaluate heat and work exchanges for heat engines and heat pumps; to describe first and second order phase transformations; and to evaluate single component phase diagrams)
2. Describe the statistical thermodynamic behavior of systems (e.g., to determine the thermodynamic probabilities of allowed macrostates and of assemblies for various systems; to describe the conditions for systems following Bose-Einstein, Maxwell-Boltzmann, and Fermi-Dirac statistics; to identify the most probable macrostate in a large, many-particle system; to understand the derivation of the Boltzmann equation; and to evaluate the configurational entropy for a randomly-mixed/ideal solution)
3. Utilize the 1st, 2nd, and 3rd laws of thermodynamics for open systems, along with the use of Maxwell's equations and auxiliary functions, to understand equilibrium associated with multicomponent systems (e.g., to describe the behavior of ideal and non-ideal gas mixtures; to describe the behavior of ideal and non-ideal liquid and solid solutions; to evaluate multiphase equilibrium in binary and higher-order systems; to correlate plots of the molar Gibbs free energies of various phases vs. composition to binary constitution phase diagrams, including for systems with eutectic, peritectic, and other invariant reactions)
4. Describe the equilibria of systems undergoing chemical reactions (e.g., to develop alternate statements of equilibrium associated with chemical reactions; to determine the equilibrium state of a mixture of reactants capable of undergoing one or more chemical reactions; to understand the construction and use of Ellingham, Kellogg/Phase Stability, and Boudouard diagrams)

General Administrative Matters:

Mental Health/Wellness:

Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette.

General Administrative Matters (continued):

Campus Emergency Policy:

Information on emergency preparedness at Purdue is available at http://www.purdue.edu/emergency_preparedness. More specific information on emergency procedures for Armstrong Hall (ARMS) can be found at: https://www.purdue.edu/ehps/emergency-preparedness/emergency-plans/bep/building-beps/arms-bep.html This information includes procedures for a fire alarm, tornado warning, and active threat (such as a shooting). In the event of a fire alarm, immediately evacuate ARMS and proceed to the Engineering Mall away from the building. If a campus tornado warning has been issued, move to the lowest level possible away from exterior doors and windows. Seek more information on storm conditions from National Weather Service weather radio or application on a mobile device. In the event of an active threat, and if one cannot get away from ARMS, shelter in a room that is securable preferably without windows. In the event of a major campus emergency, course requirements and grading are subject to changes.

Electronic Devices:

Use of cell phones and similar devices, including texting, is not allowed during class. However, please make sure that such devices are set to silent or vibrate mode in order to be informed in case of a campus emergency.

As discussed above, calculators, cell phones, laptops, smart watches, or other electronic devices (e.g., devices for electronic storage, communication, and/or internet access) are not allowed to be used during quizzes or exams.

Academic Dishonesty:

Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty."

Before starting any quiz or exam, students will be required to sign the following Purdue Honor Pledge on the front page of the quiz or exam:

Purdue Honor Pledge

"As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together — We Are Purdue."

By signing below, I acknowledge that I will obey the Purdue Honor Pledge above during this closed Quiz or Exam. This includes (but is not limited to) not using a calculator, cell phone, laptop, smart watch, or any other electronic device (e.g., electronic storage or internet-accessible devices) to help answer questions on this quiz or exam. This includes (but is not limited to) not looking at any notes or reference materials or applications (such as ChatGPT) during this quiz or exam. This includes answering the questions on this quiz or exam by myself.

Name (signature): _____

There are many dishonest ways to gain an advantage over another student in an assignment (including a quiz or exam). Rules cover any assignment for which the instructor assigns a grade.

The unauthorized reproduction of class notes, teaching material, quizzes, exams, and associated solutions (i.e., reproduction other than for your personal use for this semester) is not allowed. No filming of lectures is authorized.

Helpful Tips for Succeeding in this Course:

1) **Ask questions!** If you don't understand something, the odds are that you are not alone. For most of us (including me), concepts in thermodynamics are not always intuitive. Much of the understanding in thermodynamics comes about by pursuing answers to questions (in class or during office hours).

2) **Keep up with the lectures and reading assignments.** If you keep on pace with the lectures (e.g., *after a given class session, review the lecture that night*), reading, and homework assignments, then you can ask questions (in class or during office hours) as a given topic area is covered (or shortly thereafter). This is a much more effective approach for learning thermodynamics than waiting until the last minute before an exam to get questions answered (the latter "cramming" approach is a time-tested strategy for poor performance in this class).

3) **Do the homework in an independent and timely fashion**. You will learn far more by taking a stab at solving homework problems on your own *before seeing the solutions or before asking for help*, than if you simply look at the solutions or obtain help without attempting the problems first (i.e., it helps to "know what you don't know" as you start solving a problem, so that a subsequent explanation is more valuable and memorable). While the homework problems will not be graded, the material covered in the homework problems can be very important (big hint here) for quizzes and exams.

4) **Attend office hours**. Office hours will be devoted to answering your questions regarding homework problems (although questions from reading assignments and lecture material are also welcome). Since the solutions to homework problems will be provided on Brightspace, it is expected that you will come to office hours with specific questions about these homework problems and solutions.

Overview of Class Topics

(Chapters 1-13 of D.R. Gaskell, D.E. Laughlin, *Introduction to the Thermodynamics of Materials* 2024; select sections of C.H.P. Lupis, *Chemical Thermodynamics of Materials,* 1983; and F.W. Sears, G.L. Salinger, *Thermodynamics, Kinetic Theory, and Statistical Thermodynamics,* 1984)

- Introduction
  - Applications and Limitations of Materials Thermodynamics
  - Definitions of Thermodynamics Terms
- The First, Second, and Third Laws of Thermodynamics (Closed Systems)
  - Statements of the First Law
  - Enthalpy
  - Heat Capacity
  - Statements of the Second Law
  - Applications of the First and Second Laws (Heat Engines, Carnot Cycle, Principle of Kelvin and Planck, Heat Pumps, Principle of Clausius)
  - Auxiliary Functions (Gibbs free energy, Helmholtz free energy)
  - Maxwell's Relations
  - Statements of the Third Law
  - Reference States
  - Kirchhoff's Law/Loops
  - Criteria for Equilibrium
  - Gibbs-Helmholtz Equation
  - First and Second Order Phase Transformations
  - Triple Point, Critical Point
  - Clausius-Clapeyron Equation
- Thermodynamics of Open Systems
  - Chemical Potential (and additional Maxwell's relations)
  - Molar Properties, Partial Molar Properties
  - Gibbs-Duhem Equations
  - Non-Ideal Gases, Fugacity, Ideal Solutions
  - Activity, Raoultian/Henrian Reference States
  - Raoult's, Henry's Laws
  - Sievert's Law
  - Equilibrium in a Heterogeneous, Multicomponent System
  - Gibbs Phase Rule
  - Graphical Descriptions of Partial Molar Properties and Chemical Equilibrium
  - Mixing Functions (G^M_m, G^M_i; H^M_m, H^M_i; S^M_m, S^M_i)
  - Ideal Solution
  - Excess Mixing Functions (G^E_m, G^E_i; H^E_m, H^E_i; S^E_m, S^E_i) and Non-Ideal Solutions

Overview of Class Topics (continued)

- Phase Diagrams
  - G_m vs. X_B Plots
  - Common Tangent Construction, Lever Rule
  - Binary Phase Diagram for Ideal Liquid and Ideal Solid Solutions
  - Solidus and Liquidus Curves
  - Binary Phase Diagrams for Ideal Liquid Solution/Regular Solid Solution or Regular Liquid Solution/Ideal Solid Solution (Minima or Maxima in Solidus/Liquidus Curves)
  - Binary Eutectic Phase Diagrams
  - Binary Eutectoid, Monotectic, and Metatectic Diagrams
  - Binary Peritectic Phase Diagrams
  - Binary Peritectoid and Syntectic Diagrams
  - Metastability
  - Complex Binary Phase Diagrams (with Multiple Invariant Points, Congruently-Melting and/or Incongruently Melting Compounds, Metastability, etc.)
- Thermochemistry (Chemical Reaction Equilibria)
  - Reaction Progress Variable
  - $\Delta G_{reaction}$, $\Delta G^o_{reaction}$, Statements of Dynamic Equilibrium
  - Reaction Quotient, Q, and Equilibrium Reaction Constant, K
  - van't Hoff Equation
  - Ellingham Diagrams
  - Kellogg (Phase Stability) Diagrams
  - Boudouard Diagram
- Statistical Thermodynamics
  - Boltzmann Equation
  - Energy Levels, Energy States, Degeneracies, and Occupation Numbers
  - Macrostates and Microstates
  - Thermodynamic Probability of a Macrostate and of an Assembly
  - Fundamental Postulate of Statistical Mechanics
  - Bose-Einstein, Maxwell-Boltzmann, and Fermi-Dirac Statistics
  - Most Probable Macrostate
  - Entropy, Disorder, and Derivation of the Boltzmann Equation
  - Configurational Entropy of a Randomly-Mixed (Ideal) Binary Solution